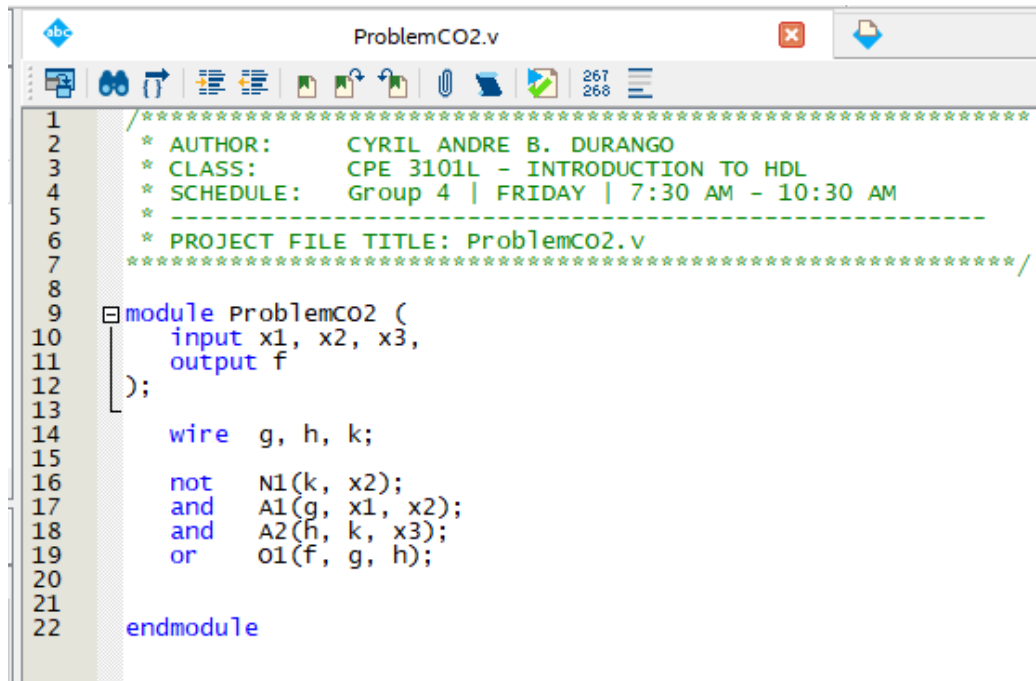


Pre-Midterm Exam for CO2



```
1  /*****  
2  * AUTHOR:      CYRIL ANDRE B. DURANGO  
3  * CLASS:       CPE 3101L - INTRODUCTION TO HDL  
4  * SCHEDULE:    Group 4 | FRIDAY | 7:30 AM - 10:30 AM  
5  *  
6  * PROJECT FILE TITLE: ProblemCO2.v  
7  *****/  
8  
9  module ProblemCO2 (  
10     input x1, x2, x3,  
11     output f  
12 );  
13  
14     wire g, h, k;  
15  
16     not    N1(k, x2);  
17     and    A1(g, x1, x2);  
18     and    A2(h, k, x3);  
19     or     o1(f, g, h);  
20  
21  
22 endmodule
```

Figure 1. Verilog Design Entry - ProblemCO2.v

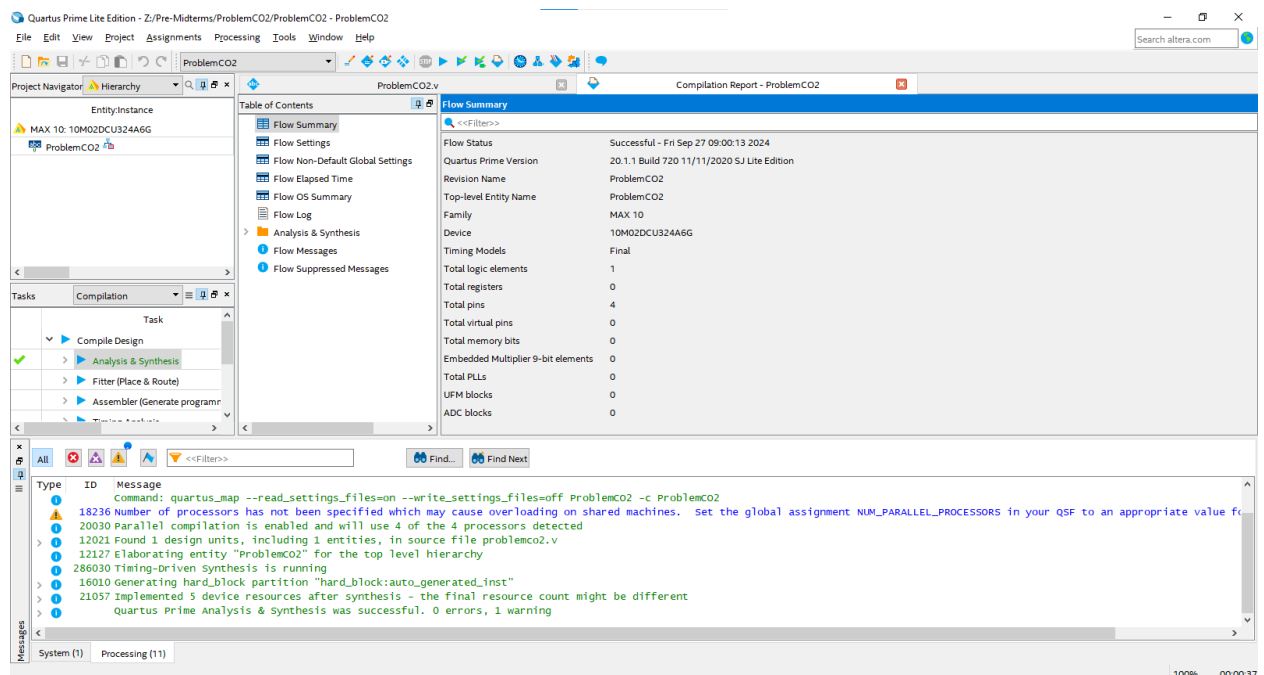


Figure 2. Compilation Report - Flow Summary of ProblemCO2.v

```

1  /*****
2  * AUTHOR:      CYRIL ANDRE B. DURANGO
3  * CLASS:       CPE 3101L - INTRODUCTION TO HDL
4  * SCHEDULE:    Group 4 | FRIDAY | 7:30 AM - 10:30 AM
5  *
6  * PROJECT FILE TITLE: tb_ProblemCO2.v
7  *****/
8
9  `timescale 1 ns / 1 ps
10 module tb_ProblemCO2 ();
11
12     reg    x1, x2, x3;
13     wire   f;
14
15     ProblemCO2 UUT (x1, x2, x3, f);
16     initial
17     begin
18
19         x1 = 0 ; x2 = 0 ; x3 = 0 ; #10
20         x1 = 0 ; x2 = 0 ; x3 = 1 ; #10
21         x1 = 0 ; x2 = 1 ; x3 = 0 ; #10
22         x1 = 0 ; x2 = 1 ; x3 = 1 ; #10
23         x1 = 1 ; x2 = 0 ; x3 = 0 ; #10
24         x1 = 1 ; x2 = 0 ; x3 = 1 ; #10
25         x1 = 1 ; x2 = 1 ; x3 = 0 ; #10
26         x1 = 1 ; x2 = 1 ; x3 = 1 ; #10
27
28         $stop;
29     end
30 endmodule
31

```

Figure 3. Testbench File tb_ProblemCO2.v

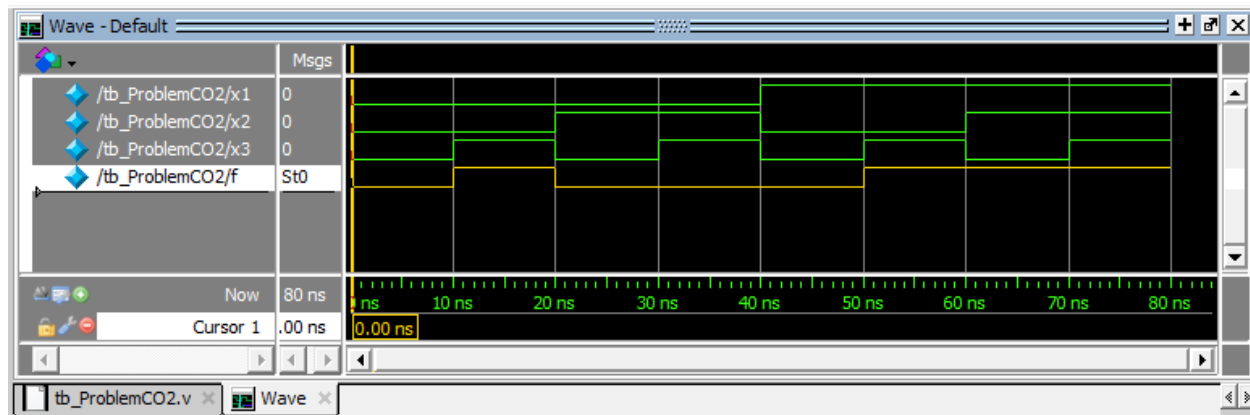


Figure 4. ModelSim Generated Simulation Waveform. Waveform in gold is the output waveform